

Computer Networks

Network Layer

The Network Layer is concerned about getting packets from source to destination, no matter how many hops it may take. It's all about routing.

Network Layer Design Issues

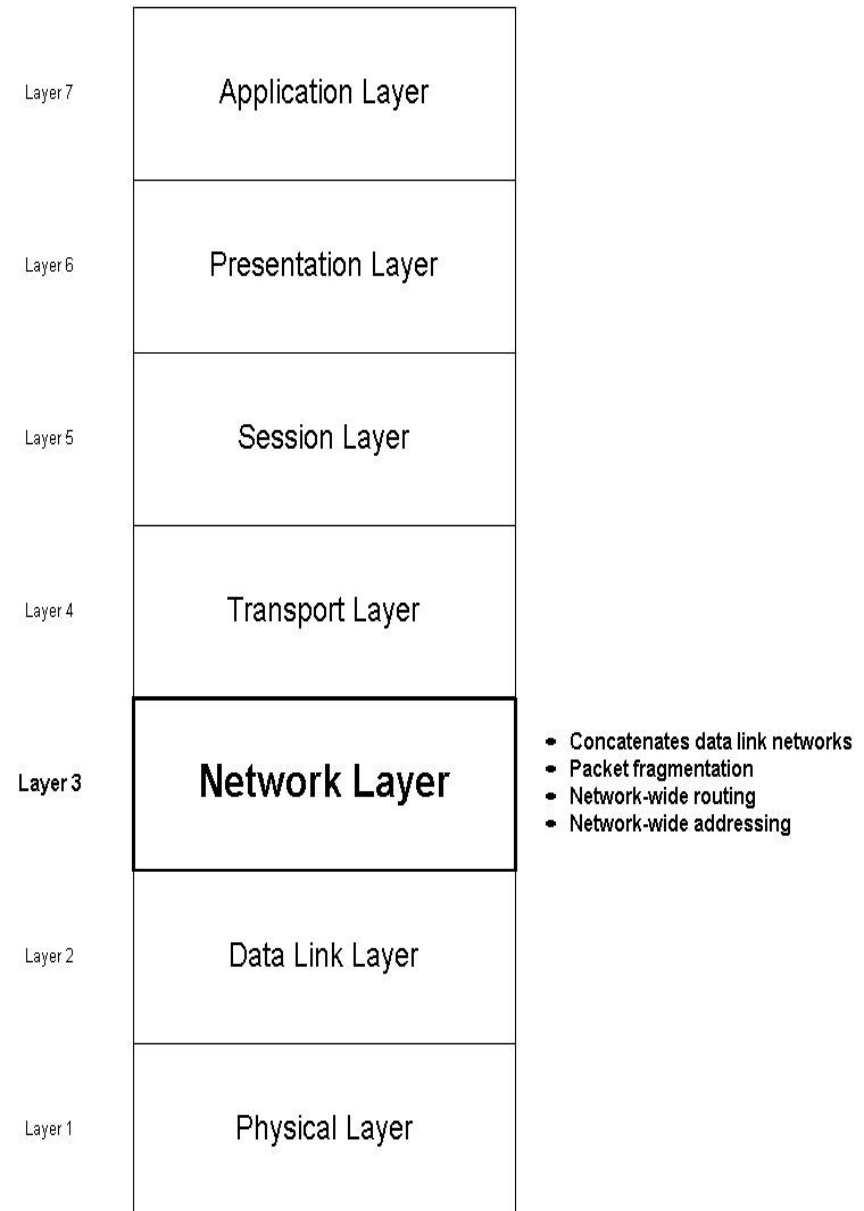
What do we need to think about in this layer?

Routing Algorithms

Strategies for getting from source to destination.

Congestion Control Algorithms

How do we keep from bottlenecking from too many packets?



Network Layer Design Issues

The network layer is responsible for routing packets from the source to destination.

- The routing algorithm is the piece of software that decides where a packet goes next (e.g., which output line, or which node on a broadcast channel).
- For connectionless networks, the routing decision is made for each datagram.
- For connection-oriented networks, the decision is made once, at circuit setup time.

ROUTING ISSUES:

The routing algorithm must deal with the following issues:

1. Correctness and simplicity: networks are never taken down; individual parts (e.g., links, routers) may fail, but the whole network should not.
2. Stability: if a link or router fails, how much time elapses before the remaining routers recognize the topology change?
3. Fairness and optimality: an inherently intractable problem.
 - a) Definition of optimality usually doesn't consider fairness.
 - b) Do we want to maximize channel usage? Minimize average delay?

CONGESTION

The network layer also must deal with congestion:

When more packets enter an area than can be processed, delays increase and performance decreases.
If the situation continues, the subnet may have no alternative but to discard packets.

If the delay increases, the sender may (incorrectly) retransmit, making a bad situation even worse.

Overall, performance degrades because the network is using (wasting) resources processing packets that eventually get discarded.

INTERNETWORKING:

When we consider internetworking, connecting different network technologies together, there are the same problems, only worse:

1. Packets may travel through many different networks
2. Each network may have a different frame format
3. Some networks may be connectionless, other connection oriented.

Services Provided to the Transport Layer

Review of Definitions:

Connection-Oriented Service: The subnet, with the help of the network layer, should provide the following operations:

1. The sending side of the pair should open a connection with its peer.
2. This connection has properties negotiated by the pair.
3. Communication is bi-directional and packets are delivered in order.
4. Flow control is accomplished in the subnet.

Connectionless Service: The subnet has no state information. It does only send_packet and receive_packet. Error control and flow control are done by the host (network or higher layers.)

Virtual Circuit: Avoids choosing a new route for each packet. A virtual circuit is a state -- it remembers how to send a packet from source to destination. This state is held in the subnet.

Datagrams: Each packet sent is routed independently through subnet. Decisions are made with the help of routing table, however this method is more robust.

Routing Algorithms

Routing algorithms is a part of network layer software which is responsible for deciding output line for incoming packets.

A router has two processes inside it. One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing tables. The other process is responsible for filling in and updating the routing tables.

There are two types of routing algorithms:

Nonadaptive algorithms - routes never change once initial routes have been selected. Also called static routing.

Adaptive algorithms – Change their routing decisions by using dynamic information as current topology, load, delay, etc.

Either of these algorithms can be applied to either datagrams or virtual circuits.

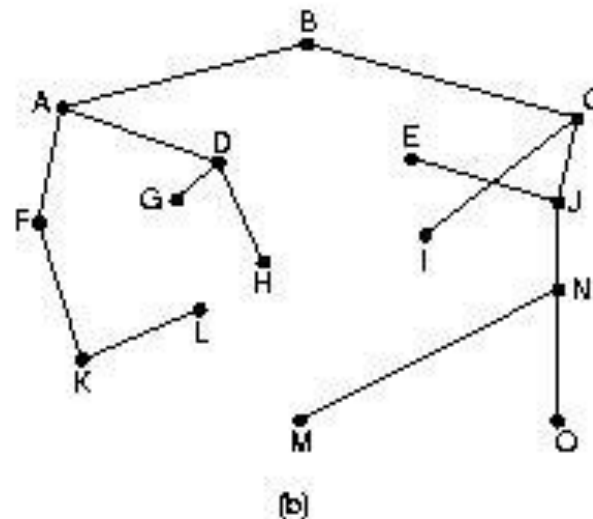
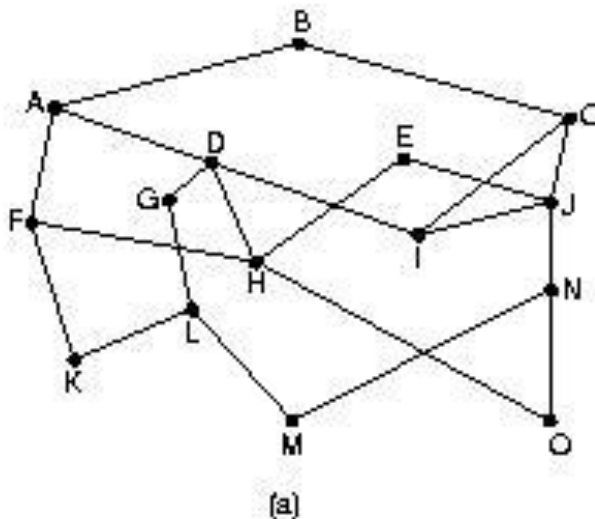
Obviously, adaptive algorithms are more interesting, as nonadaptive algorithms don't even make an attempt to handle failed links.

Optimality Principle

This simply states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along this same path.

Let call the part of the route from I to J r_1 and the rest of the route r_2 . If a route better than r_2 existed from J to K, it could be concatenated with r_1 to improve the route from I to K.

This means we can form a sink tree : the set of optimal path routes from all sources to a given destination form a tree rooted at the destination.



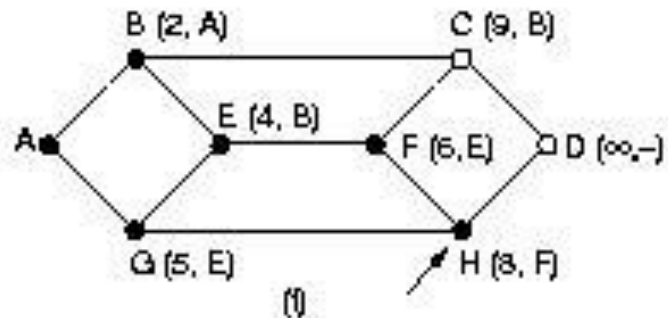
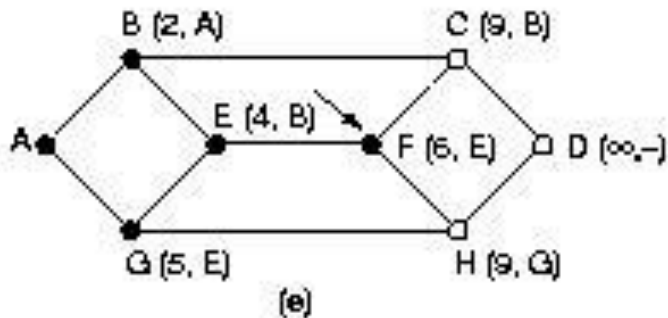
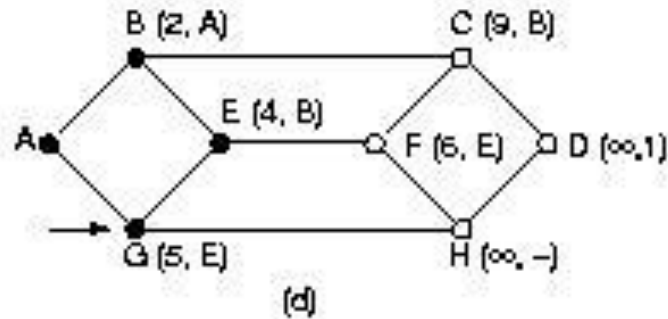
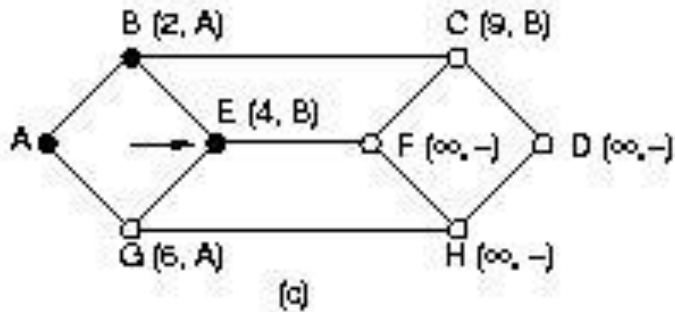
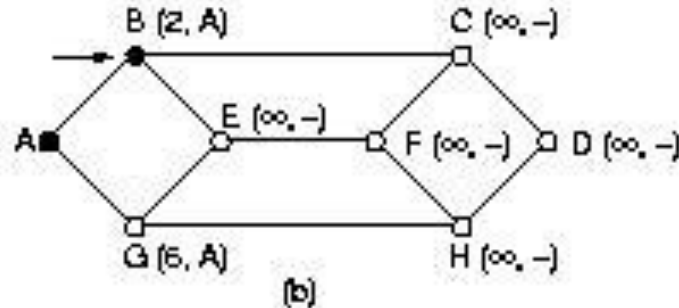
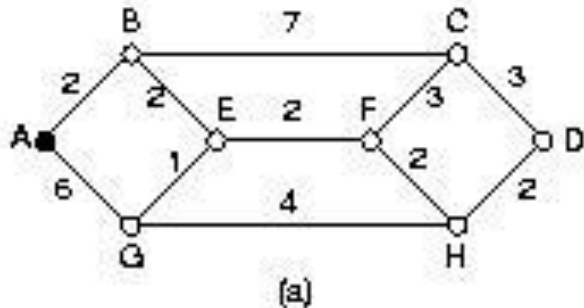
Shortest Path Routing – Dijkstra's algorithm

- Each node of graph representing router and each arc of the graph representing a communication line.
- Each node is labeled with its distance from the source node along the best known path.
- Initially, no paths are known, so all nodes are labeled with infinity.
- As the algorithm preceded and paths are found, the label may change.
- A label may be either tentative or permanent. Initially all labels are tentative.
- When it discovered that a label represents the shortest possible path from the source to that node, it is made permanent and never changed thereafter.

**Non-Adaptive
Algorithm**

Shortest Path Routing

We want to find the shortest path from A to D.



Algorithm
result
s:

nodes

(A, 0, -)
(B, 2, A)
(C, 9, B)
(D, 10, H)
(E, 4, B)
(F, 6, E)
(G, 5, E)
(H, 8, F)
(D, 10, -)

A packet is sent by a source node to every one of its neighbors. At each node, an incoming packet is retransmitted on all outgoing links except for the link on which it arrived.

To prevent packets from looping forever,

- each packet contained a hop count filed in the packet header.
- The count can originally set to no of hopes from source to destination.
- Each router decrements a hop count by one when a packet passes through it
- Whenever the hop count decrements to zero, the router discards the packet.

To reduce looping even further:

1. Add a sequence number to each packet's header.
2. Each router maintains a private sequence number. When it sends a new packet, it copies the sequence number into the packet, and increments its private sequence number.
3. For each source router S, a router:
 - a) Keeps track of the highest sequence number seen from S.
 - b) Whenever it receives a packet from S containing a sequence number lower than the one stored in its table, it discards the packet.
 - c) Otherwise, it updates the entry for S and forwards the packet.

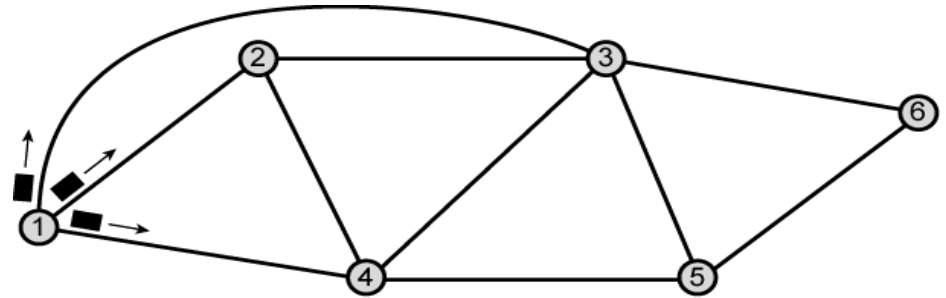
Example:

A packet is to be sent from node 1 to node 6 and is assigned a hope count of 3.

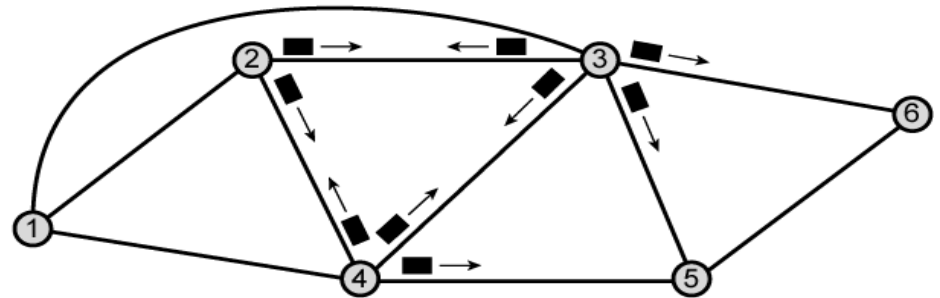
On first hope, three copies of a packet are created and hope count decrement by one.

On second hope of all these copies, a total of nine copies are created. One of these copy reaches at node 6.

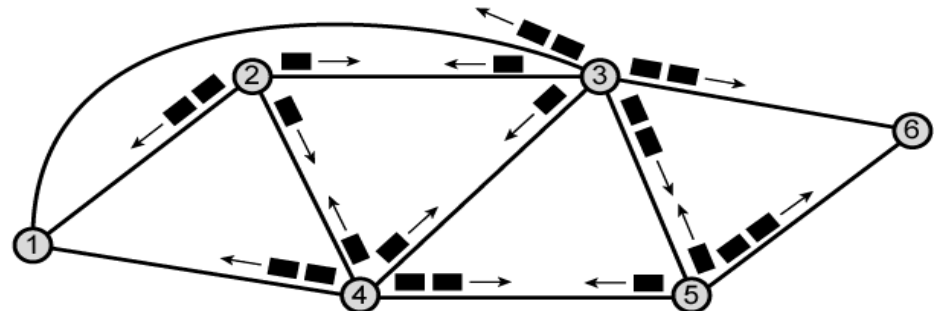
On third hope, 22 copies are generated.



(a) First hop



(b) Second hop



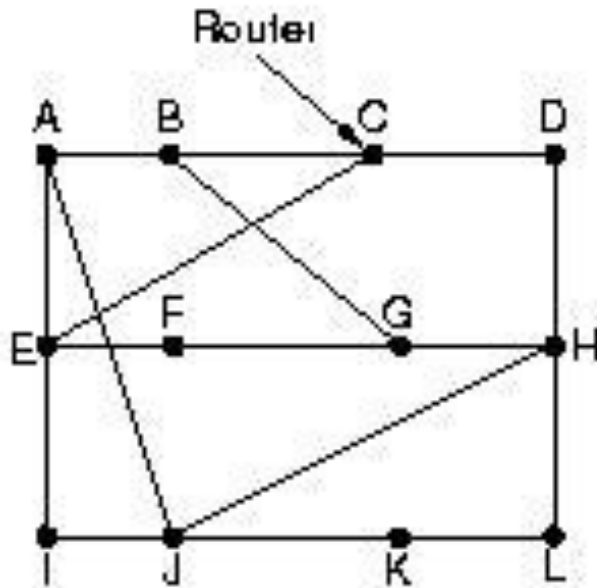
(c) Third hop

Distributed Bellman-Ford and Ford-Fulkerson routing algorithm:

Routers work together.

1. Each router maintains a routing table (vector). It is containing one entry for each routers. This entry contains two parts: preferred outgoing line to use for that destination and an estimate of time (delay). Tables are updated by exchanging information with neighbors.
2. Each router knows the delay of reaching its neighbors (e.g. send echo requests).
3. Routers periodically exchange routing tables (delay vectors) with each of their neighbors.
4. Upon receipt of an update, for each destination in its table, a router:
 - Compares the delay in its local table with the delay in the neighbor's table plus the cost of reaching that neighbor.
 - if the path via the neighbor has a lower cost, the router updates its local table to forward packets to the neighbor.

Distance Vector Routing



This algorithm was used in the original ARPANET. Unfortunately, it suffers from the problem: good news travels quickly, bad news travels slowly (count-to-infinity problem).

To	A	I	H	K	New estimated delay from J	
					↓	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	8	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	-
K	24	22	22	0	8	K
L	29	33	9	9	15	K

JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 8

{ Vectors received from J's four neighbors }

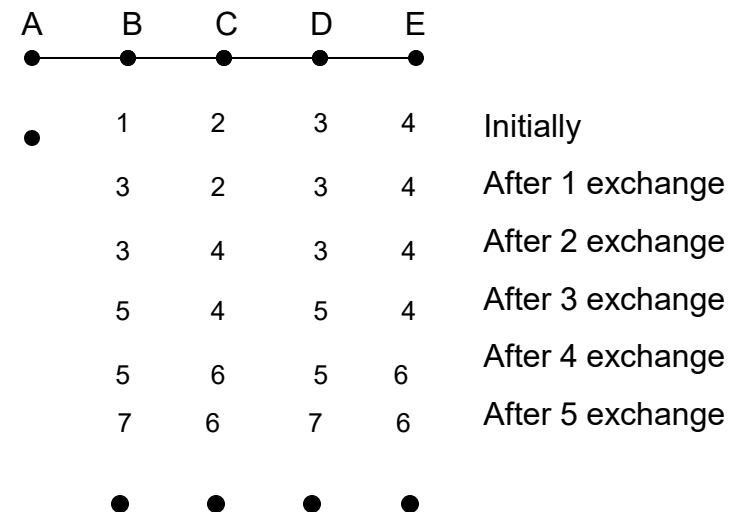
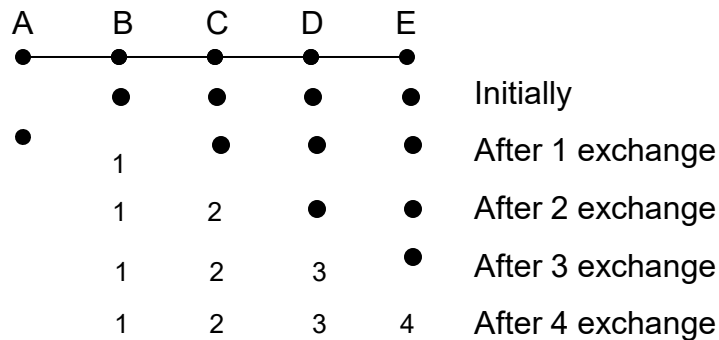
{ New routing table for J }

The Count-to-Infinity Problem

Good news travels quickly, bad news travels slowly (count-to-infinity problem).
Consider the five nodes (linear) subnet and delay metric is the number of hops.

In fig 1. Suppose A down initially, they all recorded the delay to A is infinity. When A comes up. At the first exchange...

In Fig 2. All the line and routers are initially up. Router B, C, D, and E have distances to A 1, 2, 3, and 4, respectively. Suddenly A goes down then ...



Link State Routing

In the new algorithm:

1. Each router maintains a database describing the topology and link delays between each router. That is, each router keeps track of the full graph of links and nodes.
2. Each router periodically discovers its neighbors
 - **Learning about the neighbors:** Sends a special HELLO packet on booting and receiver router send back reply telling their networks addresses.
 - **Measures the delays across its links:** Send a special ECHO packet on the line and the receiving routers send back reply immediately. Measuring the round trip time and dividing it by two to get the delay.
 - **Building the link state packet :** once the information needed for exchange has been collected, each routers build a packet containing all the data. (id of sender followed by a sequence number and a list of neighbors and their delays.
 - **Distributes the link state packet:** Forwards that information to all other routers.
3. Avoids the count to infinity problem since all routers get each other router's information.
4. Updates are propagated at high priority using flooding.
 - Updates contain sequence numbers, and a router forwards 'new' copies of the packet.
 - Why use flooding? Because that way routing updates propagate even when routing tables aren't quite correct.
 - ACKs are sent to neighbors.
5. **Compute the new route:** Each router uses an SPF algorithm to calculate shortest paths based on the current values in its database.
6. Because each router makes its calculation using the same information, better routing decisions are made.

Hierarchical Routing

One of the fundamental issues regarding routing is scaling.

- a) As a network becomes larger, the amount of information that must be propagated increases, and the routing calculation becomes increasingly expensive.
- b) Obviously, there are limits to how big a network can be.

Hierarchical routing is an approach that hides information from far-away nodes, reducing the amount of information a given router needs to perform routing:

Divide the network into regions, with a router only knowing the details of how to route to other routers in its region.

- a) In particular, a router does not know about the internal topology of other regions.
- b) Gateway is a router that knows about other regions.

A node in each region is designated as an entry point, and the entry point knows how to reach the entry points in all the other regions.

When traffic flows from A to B, it actually follows the path

A - AENTRY - BENTRY - B,

where AENTRY and BENTRY are the entry points to the respective regions.

Hierarchical Routing

Advantage: Scaling. Each router needs less information (table space) to perform routing.

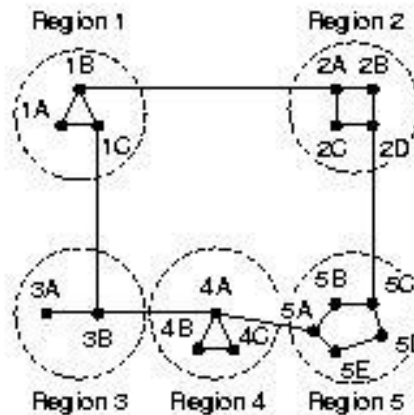
Disadvantage: Sub optimal routes. The average path length increases because there may be a shorter path that bypasses the entry points, but we don't use it.

See the Figure.

Hierarchical routing can be extended to multi-levels.

Example: telephone system:

- Area code identifies a region.
- Area code plus the first three digits identify the central office within a specific region.



Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Congestion Control

When too many packets are present in the subnet, performance degrades.

CONGESTION MECHANISMS:

When one part of the subnet (e.g. one or more routers in an area) becomes overloaded, congestion results. Congestion can be dealt with by:

1. Congestion control - when the problem occurs, limit senders or reroute. This is a reactive mode.
2. Congestion prevention - make resources available and enforce good behavior so that the congestion won't occur.

Note these definitions:

Congestion control means making sure the subnet can handle the offered traffic.

Flow control means preventing one sender from overflowing one receiver. Congestion deals with wires and routers, while flow deals with hosts.

Congestion Control

CONGESTION CONTROL POLICIES:

The approach followed will include:

1. Monitor the system to detect when and where congestion occurs.
2. Pass this information to where something can be done about it.
3. Adjust the system. This can be done by:
 - The subnet must prevent additional packets from entering the congested region until those already present can be processed.
 - The congested routers can discard queued packets to make room for those that are arriving.

Layer	Policies
Transport	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy • Timeout determination
Network	• Virtual circuits versus datagram inside the subnet • Packet queueing and service policy • Packet discard policy • Routing algorithm • Packet lifetime management
Data link	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy

See Figure for mechanisms that affect congestion.

Congestion Control

Congestion Prevention Policies

Pre-allocation schemes aim to prevent congestion from happening in the first place.

- For example, we can require that resources be pre-allocated before any packets can be sent, guaranteeing that resources will be available to process each packet.

In **virtual circuit networks**, for example, the sender opens a connection before sending data.

- The circuit setup operation selects a path through the subnet, and each router on the path dedicates buffer space and bandwidth to the new circuit.

What happens when a user attempts to open a virtual circuit and the subnet is congested?

- The subnet can refuse to open the connection, forcing the user to wait until sufficient resources become available.

Note: The ability of the subnet to reject requests to open connections is an important property of connection oriented networks.

Congestion Control

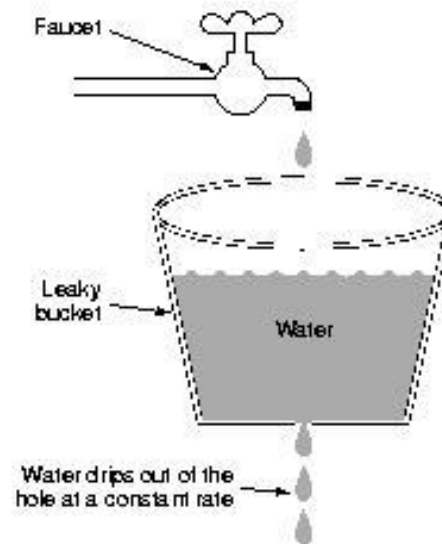
Control the rate at which packets are sent (not just how many). Widely used in ATM networks.

At set up, the sender and carrier negotiate a traffic pattern (shape).

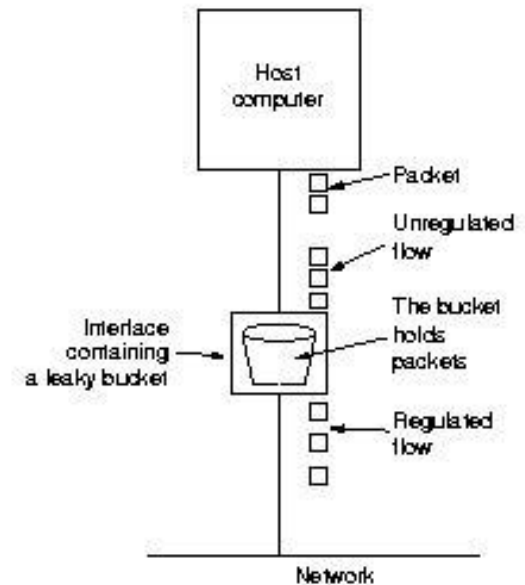
Leaky Bucket Algorithm: used to control rate in a datagram network.

- A single-server queue with constant service time.
- If bucket (buffer) overflows then packets are discarded.

Enforces a constant output rate regardless of burstiness of input. Does nothing when input is idle.



(a)



(b)

Traffic Shaping

Congestion Control

Traffic Shaping

Token Bucket Algorithm:

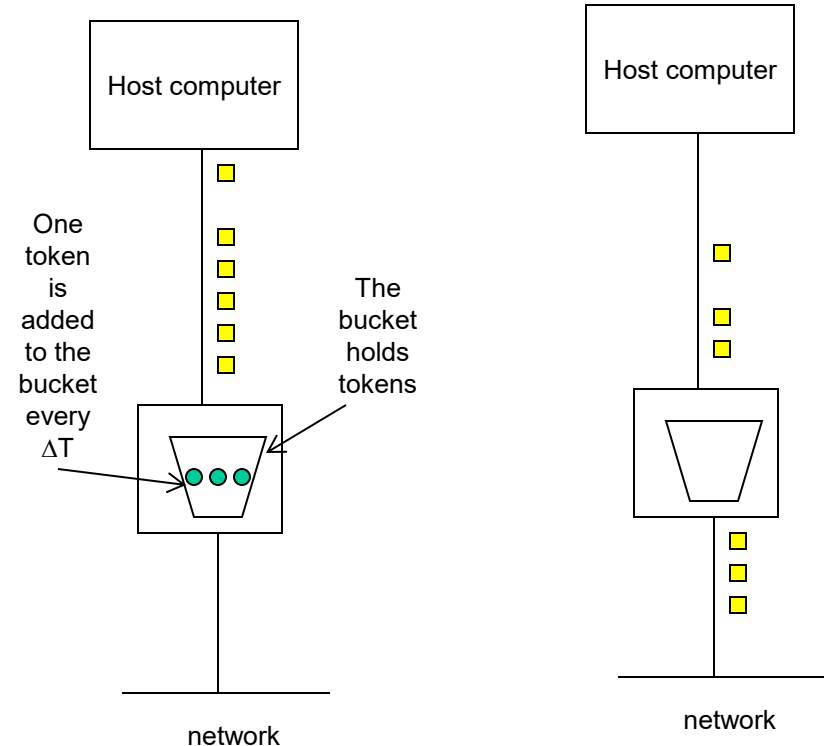
In this algorithm, A leaky bucket hold the tokens, generated by a clock at the rate of one token every ΔT sec.

For a packet to be transmitted, it must capture and destroy one token.

The leaky bucket algorithm does not allow idle hosts to save up permission to send large burst later.

The Token bucket algorithm does allow saving , up to the maximum size of the bucket, n. this mean that bursts of up to n packets can be sent at once allowing some burstiness in the output stream.

The token bucket algorithm never discard the packet when bucket fill up, while the leaky bucket algorithms discard the packet when bucket is full.



Congestion Control

Flow Control

Flow control is one way of preventing a fast sender from overwhelming a slow receiver. Flow control can be helpful at reducing congestion, but it can't really solve the congestion problem.

- If the two machines use a sliding window protocol, and the window is large, the link will become congested in a hurry.
- If the window size is small (e.g., 2 packets), the link won't become congested. Note how the window size limits the total number of packets that can be in transmission at one time.

Flow control can take place at many levels:

- User process to user process (end-to-end). Later, we'll see how TCP uses flow control at the end-to-end level.
- Host to host. For example, if multiple application connections share a single virtual circuit between two hosts.
- router to router. For example, in virtual circuits.

Routers can do the following:

1. When congestion is present, they can send choke packets to the sender that say 'slow down'. Host listens the choke packet in a specified fixed interval. After one period is expired, the host listen for more choke packets for another interval. If one arrive, line is still congested, so host reduces the flow still more and begin ignoring the choke packet.
2. How can a router measure congestion?
 - A router might estimate the level of congestion by measuring the percentage of buffers in use, line utilization, or average queue lengths.

Advantage:

1. Dynamic. Host sends as much data as it wants, the network informs it when it is sending too much.

Disadvantages:

1. Difficult to tune.
 - a) By how much should a host slow down?
 - b) The answer depends on how much traffic the host is sending, how much of the congestion it is responsible for, and the total capacity of the congested region.
 - c) Such information is not readily available in practice.
2. After receiving a choke packet....
 - a) The sending host should ignore additional choke packets for a short
 - b) Packets currently in transmission may generate additional choke packets.
 - c) How long? Depends on such dynamic network conditions as delay.

Desirable:

More attention paid to reserving resources (buffer, bandwidth, CPU cycles etc) so that chances of congestion are reduced and the quality of service is more reliable.